

Sourav Singh Katoch

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Education

National Institute of Technology, Srinagar <i>Bachelor of Technology, Computer Science and Engineering</i>	2021 – 2025 CGPA: 8.05
M.H.A.C Nagbani School, Jammu <i>Board of Secondary Education, Jammu</i>	2021 92.4%

Experience

Botlab Dynamics Pvt. Ltd. <i>Backend Developer</i>	July 2025 – Present Onsite
- Developed backend algorithms in Python for autonomous drone path planning, computing precise geographic waypoints for aerial image capture over polygonal survey areas. - Integrated a rolling shutter pipeline to correct image distortion caused by drone motion during capture, improving accuracy of 3D reconstruction from survey imagery. - Extended QGroundControl (QGC) with DJI drone compatibility using the MAVLink protocol, enabling mission planning and real-time telemetry for DJI platforms. - Integrated geospatial calculations to ensure optimal coverage, overlap, and altitude control for accurate photogrammetry outputs.	

Open Source Contributions

LLVM/Clang — Compiler Infrastructure github.com/llvm/llvm-project/pull/204891	June 2026
- Fixed a diagnostic group hierarchy bug in Clang where <code>-Wlifetime-safety-dangling-global-moved</code> was not implied by <code>-Wlifetime-safety-all</code>, adding it to the <code>LifetimeSafetyStrict</code> group in <code>DiagnosticGroups.td</code>. - Added test coverage for the <code>lifetime-safety-dangling-global-moved</code> warning in Clang's Sema test suite, reviewed and merged by LLVM maintainers.	

Projects

Ugit — Simplified Version Control System <i>Python, Tkinter, Git</i>	Sep 2024 – Present
Desktop GUI application that simplifies Git operations for beginners, replacing CLI commands with buttons for commit, push, pull, and status while executing native Git commands in the background.	

Technical Skills

Languages	C++, Python, JavaScript
Web & Frameworks	React, Next.js, Node.js, Express.js, FastAPI, WebSockets
Drone & Robotics	MAVLink, QGroundControl, geospatial data, photogrammetry, rolling shutter correction
Tools	Git, GitHub, VSCode, PyCharm, Postman, Figma, Canva
Databases	MySQL

Achievements & Certifications

- Solved **800+** problems on [LeetCode](https://leetcode.com) — strong DSA foundation.
- Python for Data Science** — Cognitive Class by IBM (May 2025)